

5G QoS ANALYSIS AND CELLULAR PERFORMANCE MONITORING

A data-driven assessment of cellular service quality, capacity and performance indicators

Project Domain	5G Communication Networks and Data Analytics
Software / Tools	Oracle SQL, SQL*Plus, Python, Pandas, Matplotlib, CSV
Method	Rule-based KPI evaluation with normalized QoS scoring
Dataset	9 cellular sites across North, Central and South regions

Report Purpose

This report documents the analysis workflow, performance findings and engineering interpretation of the supplied 5G cell dataset. The wording and structure have been independently prepared from the project information and graphs provided.

1. Executive Summary

This study evaluates the performance of nine cellular sites using a combination of network KPIs and a normalized Quality of Service score. The analysis links database-oriented extraction with Python-based processing so that raw measurements can be converted into understandable engineering results.

The main result is that performance differs substantially between sites. CELL_N_02 provides the strongest overall result in the supplied dataset, while CELL_S_03 is the principal cell requiring investigation. The findings show why throughput should not be examined in isolation: latency, load and packet loss also affect the overall service assessment.

2. Background and Motivation

A 5G network must deliver capacity and responsiveness while maintaining reliable communication. High utilization can create congestion, excessive latency can affect interactive services, and packet loss can reduce transmission reliability. A combined KPI view therefore provides a more useful picture than a single metric.

The project is designed as a compact performance-monitoring pipeline. It extracts records, prepares them for analysis, applies project-defined limits, calculates a QoS score, and presents the results visually for easier interpretation.

3. Objectives

- Assess the condition of individual 5G cells using multiple KPIs.
- Compare throughput and latency across the selected sites.
- Detect cells that show signs of high utilization or degraded service.
- Create a 0–100 QoS score for straightforward cell-to-cell comparison.
- Produce visual evidence and practical recommendations for optimization.

4. Dataset and KPIs

The dataset contains nine cells grouped into North, Central and South regions. Each record describes cell performance using load, throughput, latency, packet loss and network-status information.

KPI	Meaning	Engineering relevance
Average Load	Resource utilization	Helps identify possible congestion or limited capacity headroom.
Average Throughput	Data transfer performance	Indicates the amount of traffic the cell can deliver.
Average Latency	Network response delay	Lower delay is generally preferable for interactive services.
Average Packet Loss	Unsuccessful packet delivery	Higher loss can indicate reliability or link-quality problems.
Network Status	Operational classification	Adds context to the numerical measurements.

5. Analysis Workflow

Step	Activity
1	Retrieve the required cell-performance records using Oracle SQL.
2	Prepare the extracted information as a structured CSV dataset.
3	Load and inspect the data with Pandas.
4	Compare KPI values and apply the selected threshold rules.
5	Calculate the normalized QoS score for each cell.
6	Generate latency, throughput and QoS comparison charts.
7	Identify priority cells and formulate network recommendations.

6. QoS Evaluation Method

The scoring process is rule-based and is intended to provide a consistent comparison within this project. The score combines the selected performance indicators and is expressed on a 0–100 scale. A larger value represents a stronger overall result under the project's scoring rules.

The following limits are used to flag potentially unfavorable conditions:

Parameter	Reference value	Use in analysis
Load	60	Values above this level indicate increased utilization.
Latency	65	Values above this level indicate comparatively high delay.
Packet Loss	4	Values above this level indicate increased packet loss.

Important note: these are project-specific thresholds, not universal 5G standards. The QoS score is therefore best used for relative comparison of the supplied cells.

7. Software and Implementation

Oracle SQL supports data extraction and filtering, while CSV provides a simple hand-off between the database stage and Python analysis. Pandas is used for tabular processing and calculation, and Matplotlib is used to create the performance graphs. Together these tools provide a lightweight end-to-end workflow for network analytics.

8. Latency Analysis

The latency results span approximately the upper-40s to upper-60s. CELL_N_02 records the lowest latency at about 48.83, making it the most responsive site in this dataset. CELL_S_01 and CELL_N_03 are among the highest, at approximately 67 and 67.3 respectively. When latency approaches or crosses the selected limit, the cell deserves additional investigation.

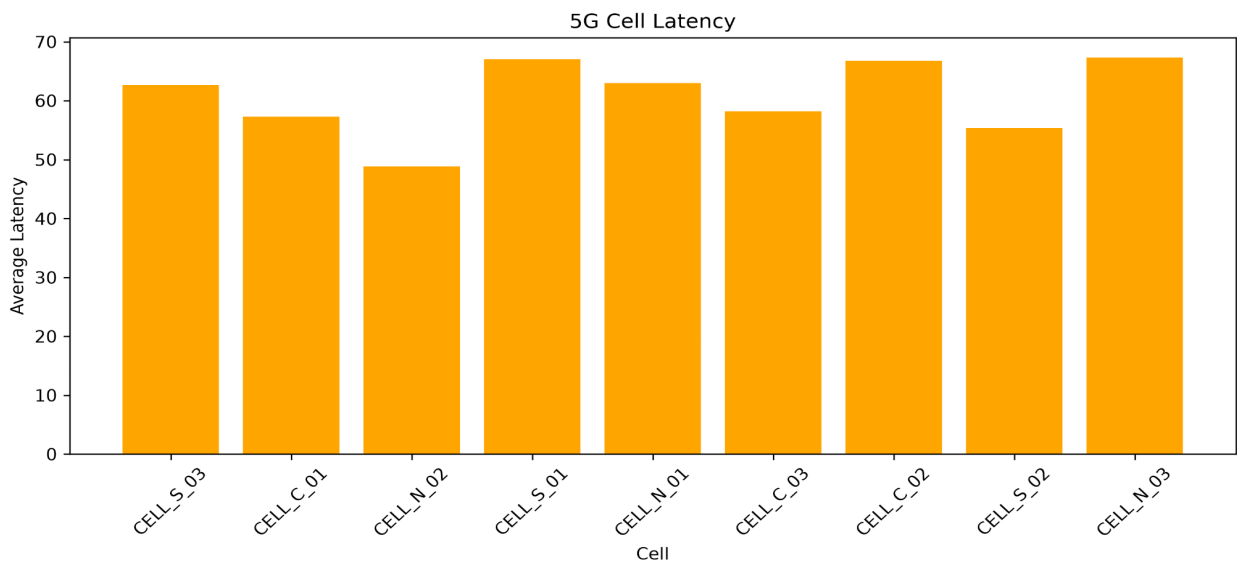


Figure 1. Average latency measured for each cellular site.

9. Throughput Analysis

Throughput is relatively strong across all nine cells. CELL_N_02 reaches approximately 104.51, the highest observed value, while CELL_C_02 is the lowest at roughly 87. The spread is moderate, which shows that throughput alone cannot explain the QoS ranking.

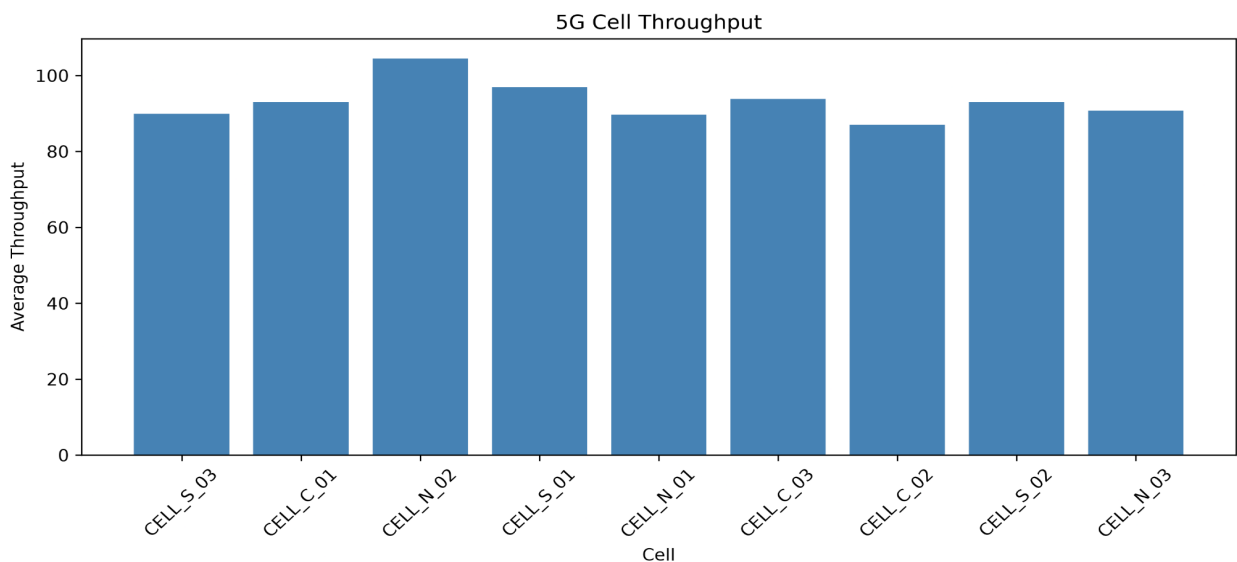


Figure 2. Average throughput by cell.

10. QoS Score Results

The normalized QoS comparison highlights a clear difference between the sites. CELL_N_02 achieves 100.0 and is the strongest overall result. CELL_S_02 and CELL_C_03 are also comparatively strong, with scores around the mid-50s. CELL_S_03 records only 14.41, making it the main candidate for corrective investigation.

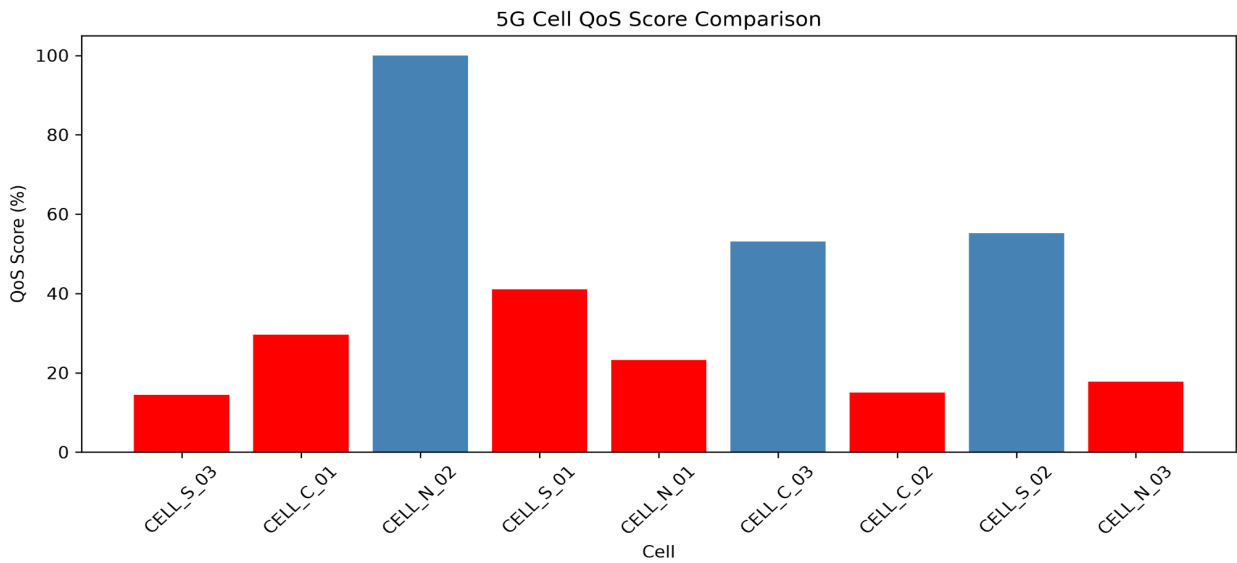


Figure 3. Normalized QoS score comparison.

11. Throughput Comparison View

The second throughput chart provides another visual comparison of the same sites. CELL_N_02 remains the leading cell, while CELL_C_02 is at the lower end. This view is useful when throughput needs to be considered alongside latency and the overall QoS score.

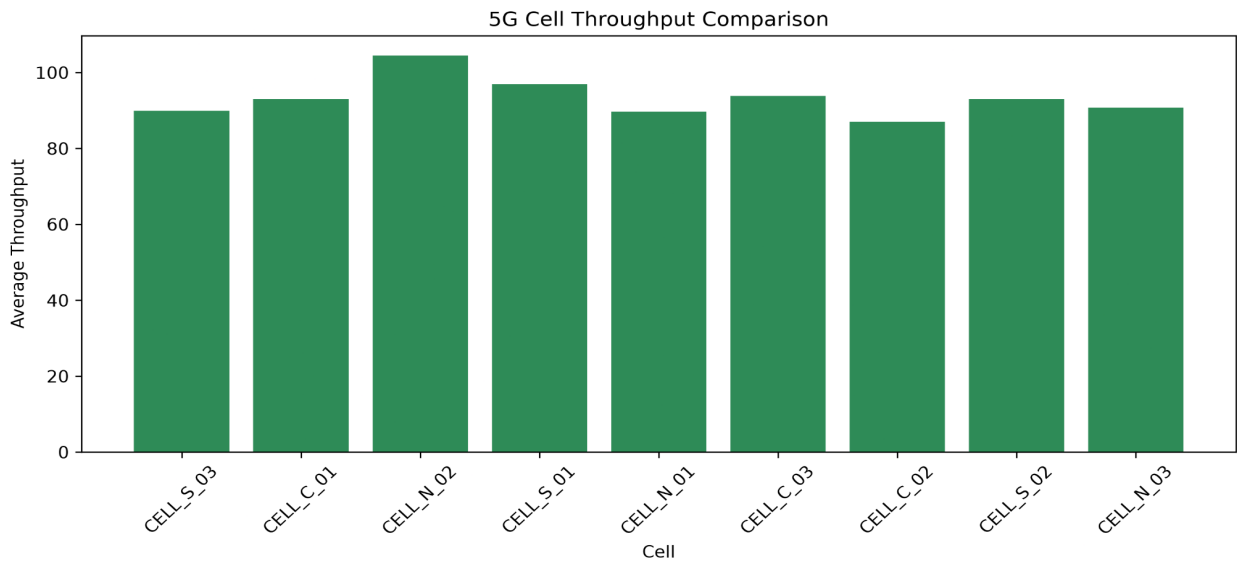


Figure 4. Comparative throughput visualization.

12. Key Cell-Level Findings

Cell	QoS	Throughput	Latency	Main interpretation
CELL_N_02	100.0	104.51	48.83	Best overall result; combines high throughput with the lowest latency.
CELL_S_03	14.41	89.90	62.70	Lowest QoS score; high-load and packet-loss conditions require attention.
CELL_S_02	≈55	≈93	≈55	One of the stronger overall QoS results.
CELL_C_03	≈53	≈94	≈58	Good throughput with moderate latency.
CELL_S_01	≈41	≈97	≈67	High throughput but comparatively high latency.

13. Recommendations

- **Prioritize CELL_S_03:** investigate its high-load condition and packet loss, then determine whether traffic balancing, radio optimization or capacity changes are required.
- **Monitor high-utilization cells:** sustained load near or above the chosen limit should trigger capacity and load-balancing checks.
- **Investigate latency:** for cells above the reference limit, examine radio-resource usage, scheduling, interference and transport/backhaul conditions.
- **Investigate packet loss:** check link quality, retransmissions, interference and transport reliability when loss exceeds the project threshold.
- **Use strong cells as references:** CELL_N_02 can serve as a performance benchmark while continuing normal monitoring.

14. Interpretation of the Results

The most important observation is that a cell can have good throughput and still receive a weak overall QoS result. This is visible in the contrast between throughput and QoS rankings. A multi-KPI assessment is therefore more informative than a capacity-only comparison.

The results should be treated as an engineering screening step. A flagged cell is a candidate for deeper investigation rather than proof of a single root cause. Actual network optimization would require additional radio, transport, traffic and time-series measurements.

15. Limitations

- The dataset contains only nine cells and represents a limited observation rather than a long-term network history.
- The QoS scoring formula and limits are project-specific and should not be presented as industry-wide standards.
- Detailed radio KPIs, user-level traffic distributions, handover statistics and time-series trends are outside the current scope.
- The graphs identify performance differences but cannot by themselves determine the exact technical cause of a problem.

16. Future Enhancements

The project can be developed into a continuous monitoring platform by adding automated database-to-dashboard pipelines, historical KPI tracking, anomaly detection, time-series forecasting and machine-learning-based QoS prediction. Interactive dashboards using Power BI or Streamlit could provide filtering by region, cell and date. Additional telecom KPIs such as handover success, radio quality and resource-block utilization could make the diagnosis more detailed.

17. Conclusion

This project demonstrates how communication-network measurements can be combined with practical data analytics to support 5G performance monitoring. The workflow moves from SQL-based extraction through Python processing and QoS scoring to visual reporting and recommendations. Within the supplied dataset, CELL_N_02 is the strongest overall performer, while CELL_S_03 deserves the highest priority for further investigation.

The work also demonstrates a useful combination of communication engineering and software/data-analysis skills. With real-time data, richer KPIs and predictive models, the same framework could be extended toward intelligent 5G network optimization.

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